



STM32MP21 product introduction

Entry level product line of the STM32MP2 series



STM32MPU Portfolio

STM32MP1 series

Industrial
communication
Processors

STM32MP133

Cortex-A7 up to 1GHz
CAN FD, 2x Eth

STM32MP153

2x Cortex-A7 + Cortex-M4
CAN FD

Analytics
and Edge
machine
vision

General
Purpose
Processors

STM32MP135

Cortex-A7 up to 1GHz
CAN FD, 2x Eth, Display, Camera

STM32MP157

2x Cortex-A7 + Cortex-M4
CAN FD, 3D GPU, DSI

STM32MP131

Cortex-A7 up to 1GHz
Eth

STM32MP151

Cortex-A7 + Cortex-M4
Eth

STM32MP2 series

STM32MP257

Dual Cortex-A35 + Cortex-M33
3x Eth/TSN (2+1 switch), 3x CAN FD

STM32MP215

Cortex-A35 + Cortex-M33
RGB LCD, MIPI CSI2

STM32MP235

Dual Cortex-A35 + Cortex-M33
H.264 Dec 3D GPU AI / NN LVDS/DSI

STM32MP255

Dual Cortex-A35 + Cortex-M33
H.264 - 3D GPU - AI / NN - LVDS/DSI

STM32MP215

Cortex-A35 + Cortex-M33
2x Eth, 2x CAN FD, RGB LCD, CSI2

STM32MP235

Dual Cortex-A35 + Cortex-M33
2x Ethernet, 2x CAN FD
H.264 Dec 3D GPU AI / NN LVDS/DSI

STM32MP257

Dual Cortex-A35 + Cortex-M33
3x Ethernet (2+1 switch) - 3x CAN FD
H.264 - 3D GPU - AI / NN - LVDS/DSI

STM32MP213

Cortex-A35 + Cortex-M33
2x Ethernet, 2x CAN FD

STM32MP233

Dual Cortex-A35 + Cortex-M33
2x Ethernet, 2x CAN FD

STM32MP253

Dual Cortex-A35 + Cortex-M33
2x Ethernet, 3x CAN FD

STM32MP211

Cortex-A35 + Cortex-M33
1x Ethernet

STM32MP231

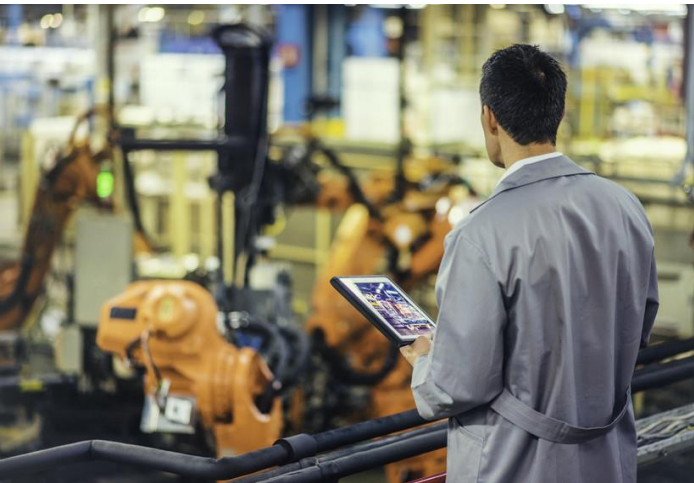
Cortex-A35 + Cortex-M33
1x Ethernet

STM32MP251

Cortex-A35 + Cortex-M33
1x Ethernet

New

STM32MP21 : Entry level to the STM32MP2 series



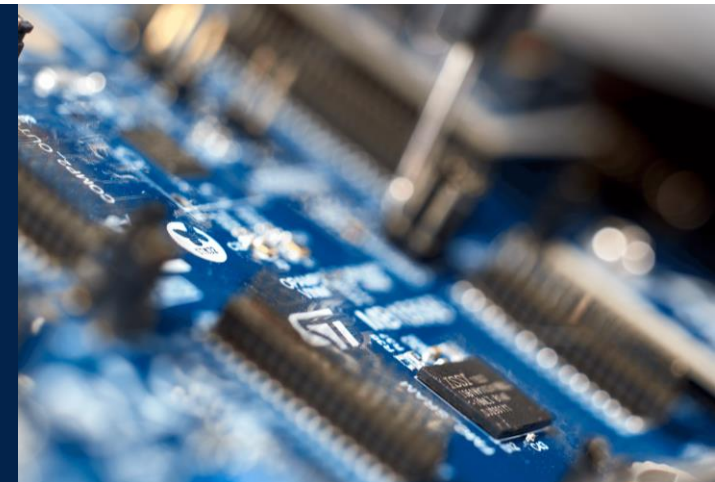
Optimize performance and power efficiency for edge devices

Ensure robust security and reliable connectivity in industrial IoT



Accelerate development and reduce time-to-market with STM32 ecosystem

Simplify design reuse and future-proof your product with MP2 series compatibility



Rich interfaces supporting the growth of connected applications



Industrial & factory automation



- Gateways
- PLCs
- HMIs
- Metering
- Bar code reader

Smart homes



- Gateways
- HMIs
- Whitegoods
- Doorbell

- Secure boot
- Firmware & data encryption
- Context isolation

Smart city and infrastructure



- Power grid
- EV charging
- Metering
- HMIs

STM32MP21 value proposition

STM32MP2 series most power efficient and cost optimized derivative

Up to 1.5 GHz & (LP)DDR4: more performance than STM32MP13

Single Cortex-A MPU, ideal for edge computing and camera use cases (CSI)



Ecosystem

(OpenSTLinux, bare metal)



Coprocessor

(*autonomous* Cortex-M)



Security target

(SESIP L3)



Robustness for complex industrial applications



Industrial qualification combining both

- 100% operation time for 10 years
- Junction temperature: - 40°C to 125°C

10-year longevity commitment renewed every year

Flexible resource allocation between cores

- Arm® Cortex®-A35 up to 1.5 GHz
- Arm® Cortex®-M33 up to 400 MHz

Advanced security for Industry 4.0





Performances & connectivity at effective cost



STM32MP2 series DNA

- Performance & real time with power efficiency
- Scalable towards STM32MP23 / STM32MP25 with pin-to-pin compatibility

Versatile connectivity

- Dual gigabit Ethernet with TSN support
- MIPI CSI interface with embedded ISP

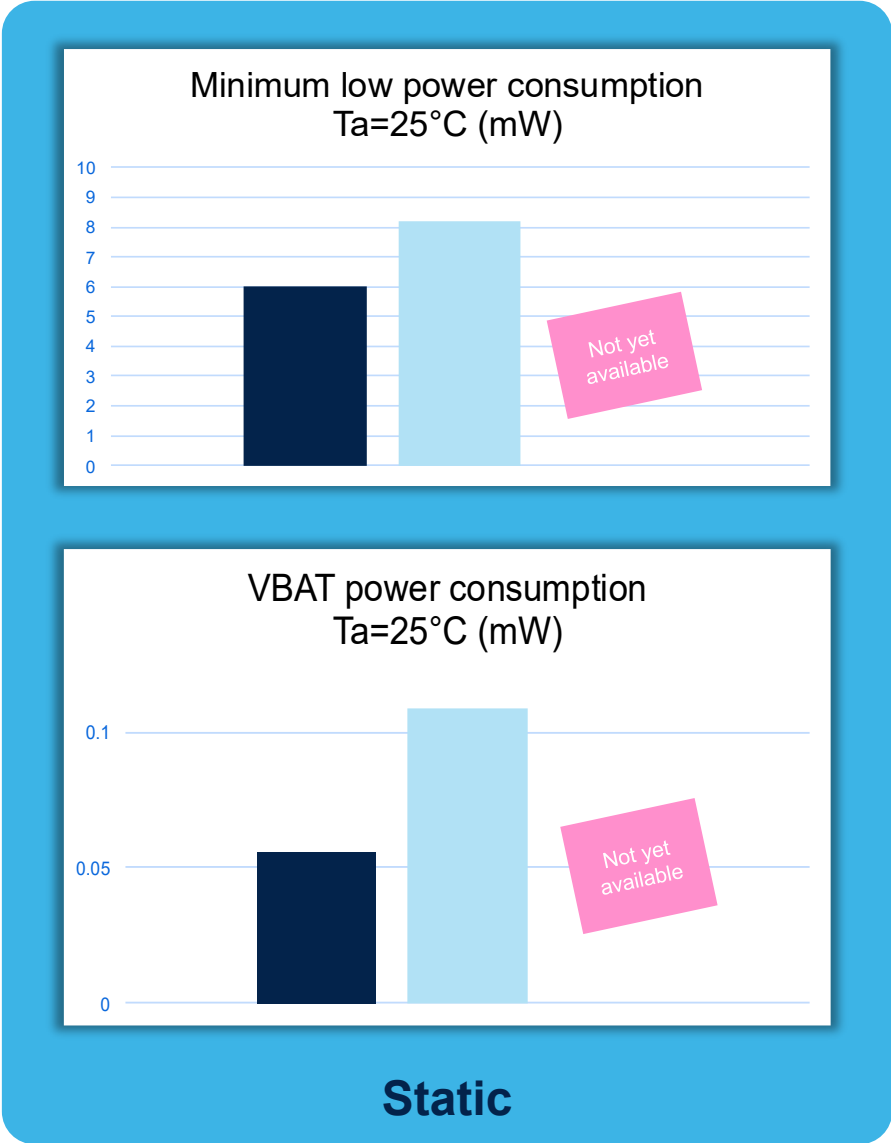
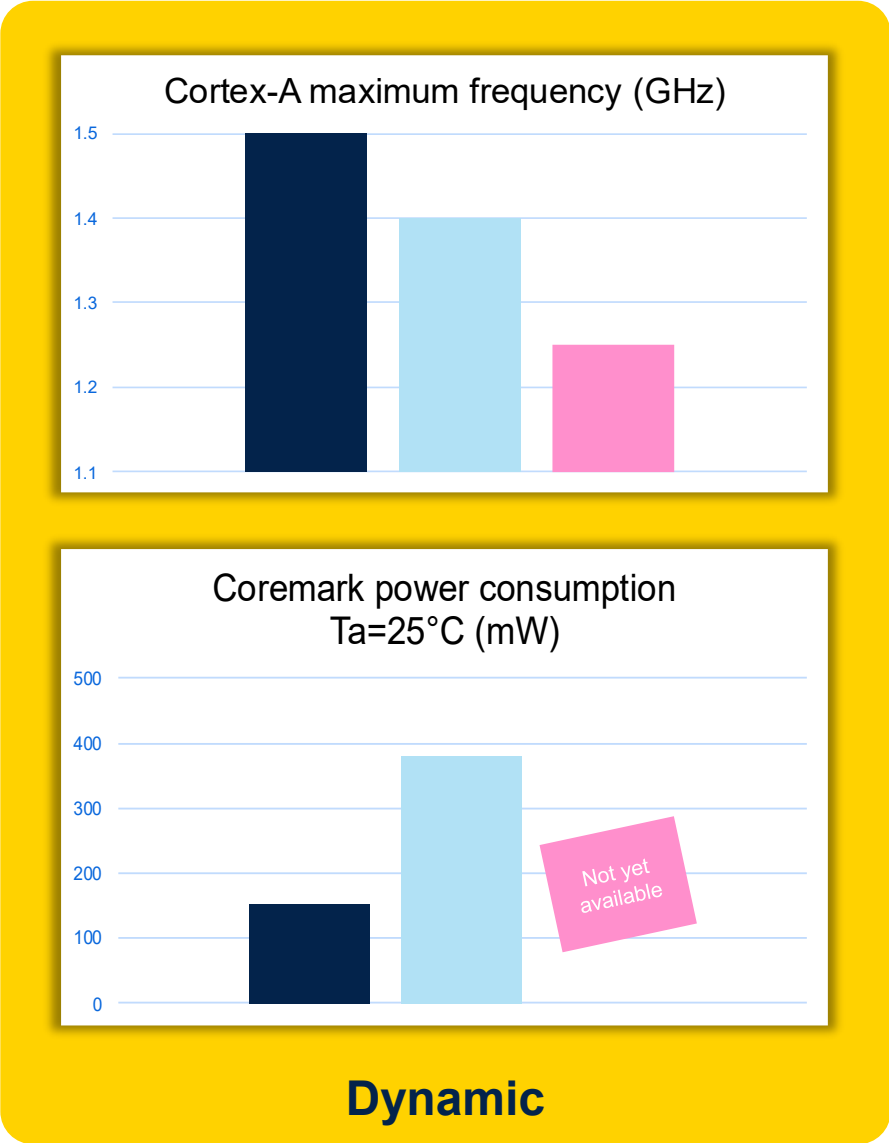
Effective cost

- Optimized architecture
- 8x8mm low-cost package





Low Power Consumption





Strong security



Memory & peripheral protections
against illegal access control



Secure cryptographic accelerator for robustness against physical attacks



Device authentication & attestation
during product life cycle



Code isolation
for runtime protection



Security assurance level 1

Software robustness



Security ecosystem

Trusted execution with OP-TEE

In-factory secure secret provisioning (SSP)

STM32Cube framework for MPU
(Signing & key generation)

and more!

Enhance your application security with STM32MP21x



CRA compliant

Resource isolation framework

Cortex-M33 as trusted boot device (M33-TD)

Enhance your application security with STM32MP21x

Navigate new EU security regulations with STM32MP21 solutions

- ✓ Build devices for the long term
- ✓ Protect users and data
- ✓ Avoid risks and penalties
- ✓ Reduce costs of non-security

How ?

- ✓ Cryptographic accelerators: AES-128/192/256, SAES with SCA, PKA with DPA protection
- ✓ 5-years maintenance in OpenSTLinux ecosystem to fit regulations requirements
- ✓ SESIP and PSA Certified target certification
- ✓ STM32Trust framework (TF-M, TF-A, OPTEE...)

CRA compliant

Resource isolation framework

Cortex-M33 as trusted boot device (M33-TD)

Enhance your application security with STM32MP21x

Isolate and manage access to hardware resources (peripherals, memory regions, and communication interfaces)

- ✓ Prevent unauthorized access to critical HW resources
- ✓ Limits the impact of SW bugs to isolated domains
- ✓ Support for multi-tenant systems
- ✓ Facilitates compliance with security standards
- ✓ Resources optimization without compromising security
- ✓ Reduced complexity and debugging efforts

CRA compliant

Resource isolation framework

Cortex-M33 as trusted boot device (M33-TD)

Enhance your application security with STM32MP21x

Leverage the Cortex®-M33 core as a trusted domain (TD) boot processor now available in OpenSTLinux distribution

- ✓ Increase security by isolating critical secure boot functions on a dedicated core
- ✓ Keep secure services on the Cortex-M33 isolated trusted domain
- ✓ Strengthen the protection of secure boot applications against attacks

CRA compliant

Resource isolation framework

Cortex-M33 as trusted boot device (M33-TD)

M33-TD: direct boot on Cortex-M

Use of the Cortex-M33 core as the Trusted Domain boot processor

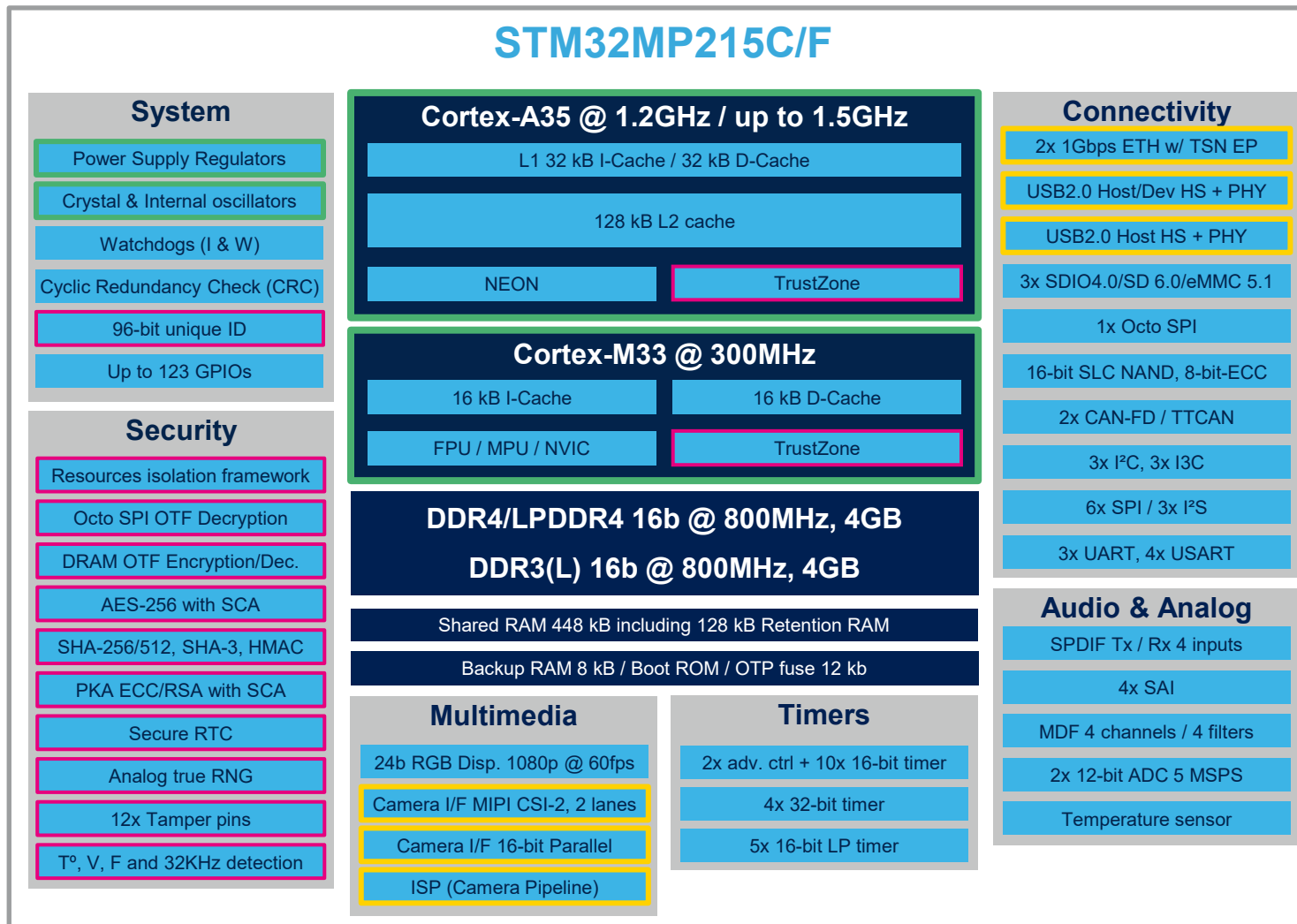
Define your needs

- ✓ You want to support Cortex-A35 independent reboot?
- ✓ Need safety?
- ✓ Need quick processing while waking up from low power modes?
- ✓ Need early services launch?
- ✓ Want dual Flash for isolated boot codes?

Benefits

- ✓ Owns security and system configurations by default
- ✓ Fast Cortex-M boot and low power exit, independent of Cortex-A
- ✓ Safety on Cortex-M, allowing Cortex-A independent reboot
- ✓ Isolated secure and system island:
 - Secure services,
 - System: clock, regulator, low power, DDR
- ✓ Dual boot: dedicated external mass storages per Cortex-M/A

STM32MP215 block diagram

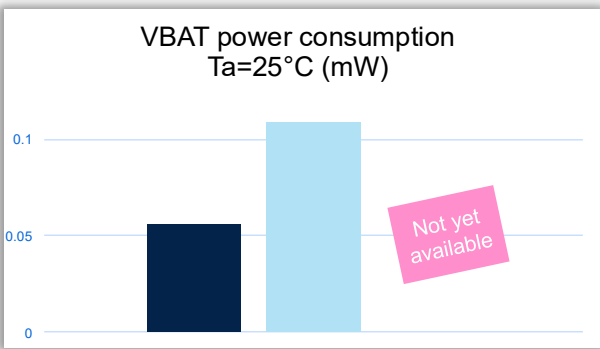
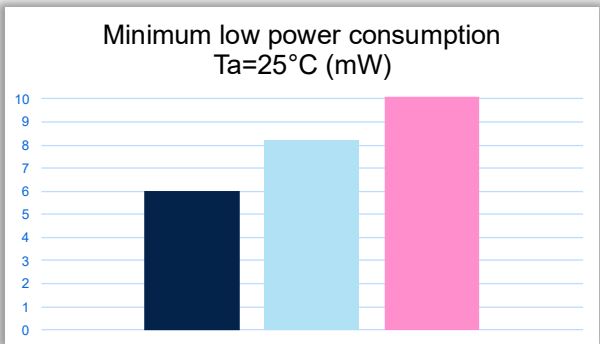
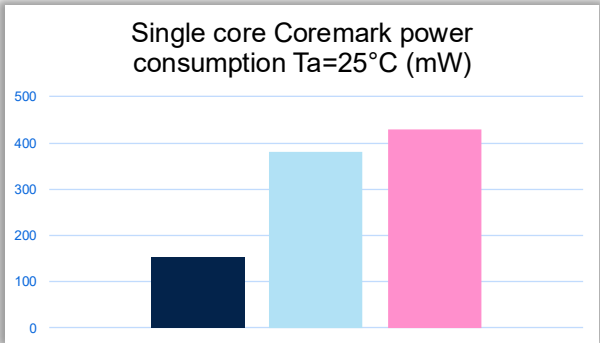
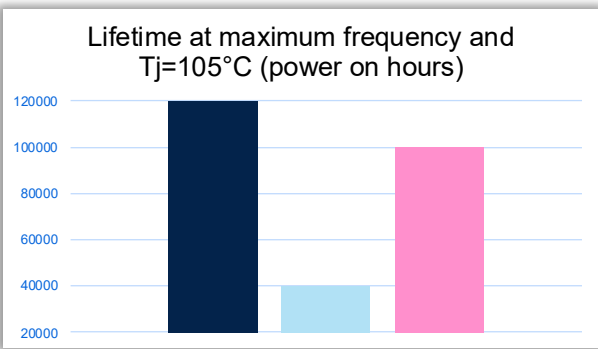
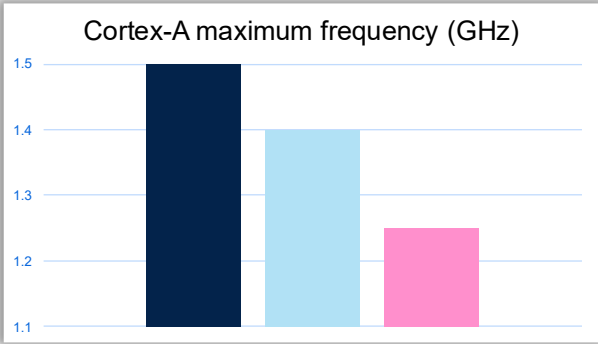
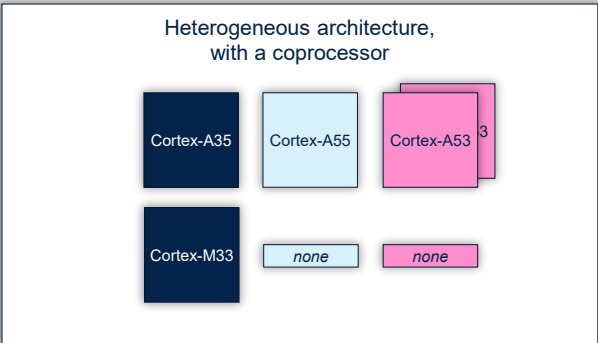
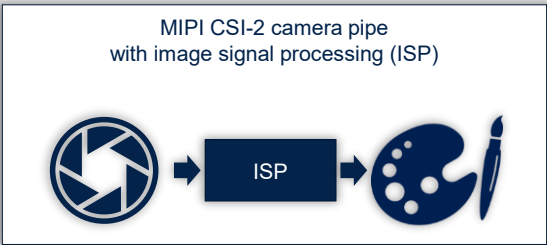
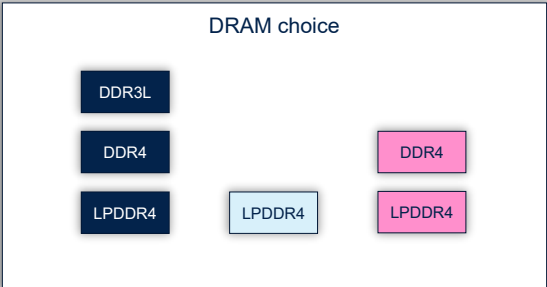
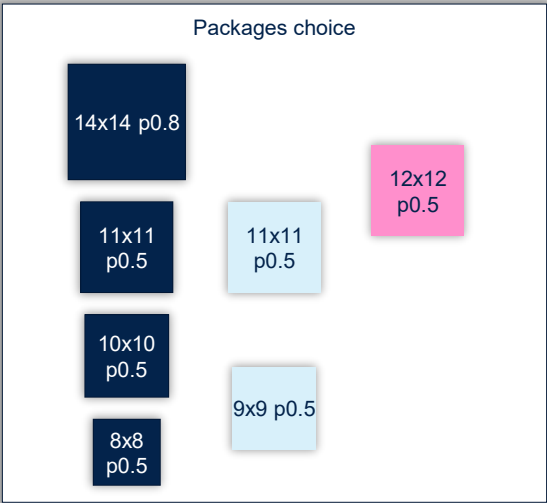


Power efficiency

Enhanced Security

Rich connectivity

STM32MP21 competitive advantages



Legend:

More dark blue is better

Less dark blue is better

Four packages



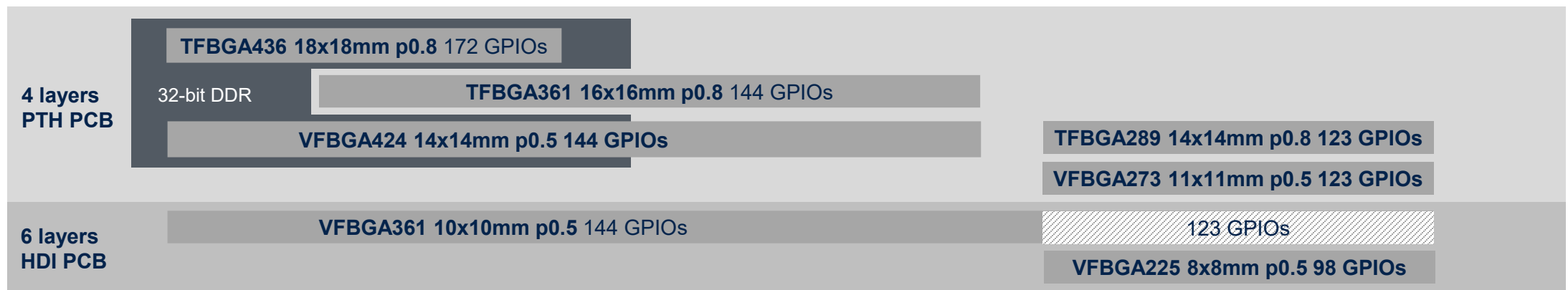
For robust and cost optimized applications

For PCB size and cost optimization

For pin compatibility with STM32MP25/23

For highly integrated MPU based systems

STM32MP2 Full Series Lineup



Pin-to-pin compatible packages and same ecosystem!

A new power management IC: STPMIC2L



QFN: 6.5x6.5 mm pitch 0.4

- **7 Buck** DC/DC converters
- **8 LDOs** + 1 DDR Reference Voltage LDO
- Provides power supplies to STM32MP2 as well as to DRAM, Flash **and other application peripherals** (USB, Display, ...)
- Better power efficiency (Buck vs. LDO)

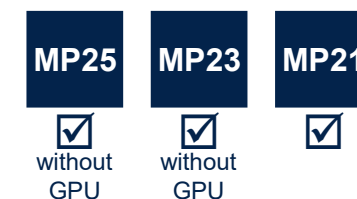
Ideal for:



QFN: 5x5 mm pitch 0.4

- **3 Buck** DC/DC converters
- **7 LDOs**
- Provides power supplies to STM32MP2 as well as to DRAM, and Flash

Ideal for:



STM32MP215F-DK Features

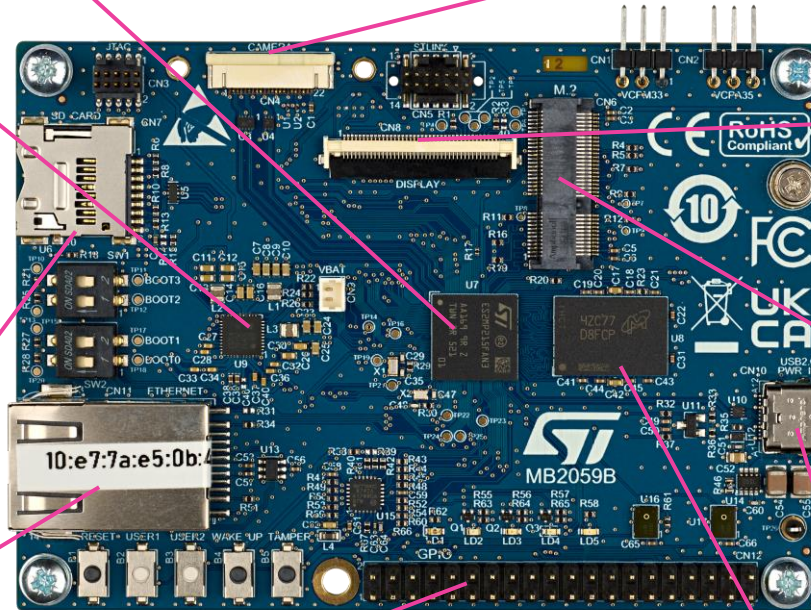
STM32MP215F VFBGA
11x11mm p0.5

STPMIC2L Power
Management IC

microSD™ card holder

Gigabit Ethernet Port with
TSN support

GPIO, Raspberry Pi
Expansion Connectors



MIPI CSI-2 2-lane Camera
Interface

LCD-TFT Display controller

M.2 connector Key E with
SDIO, UART and control lines

USB2.0 Type C connector for
power supply and data

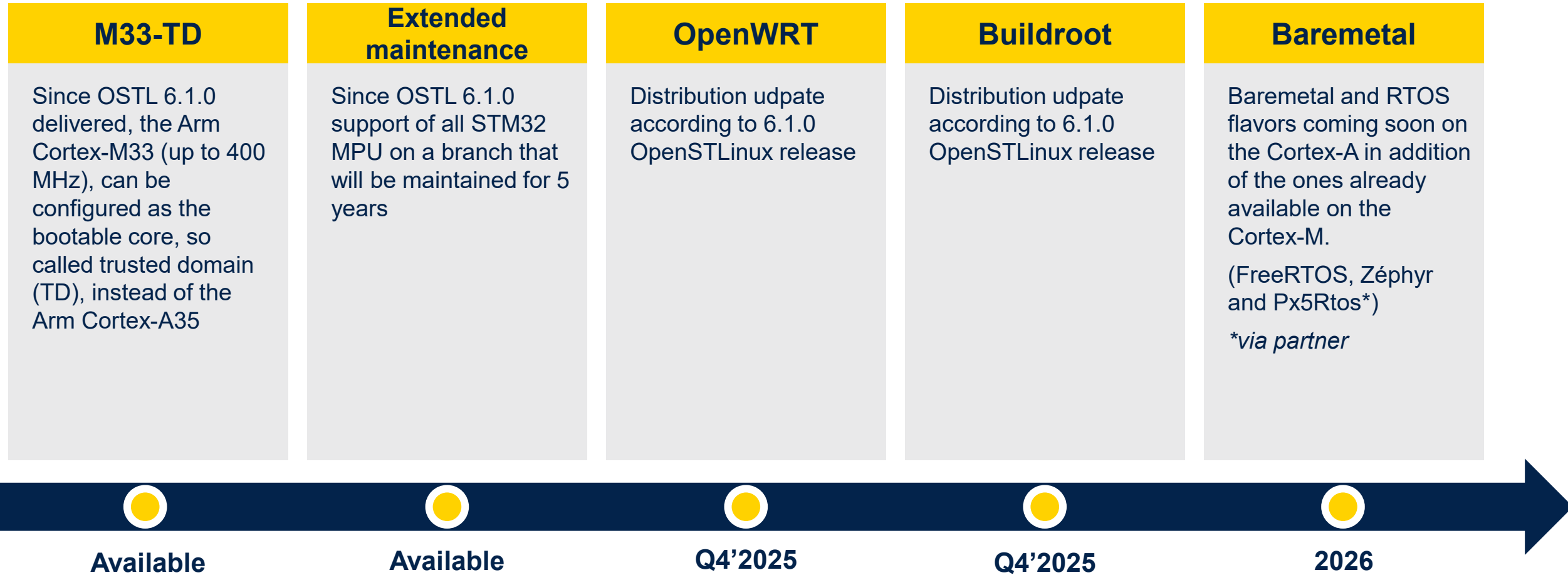
2GB LPDDR4 16-bit



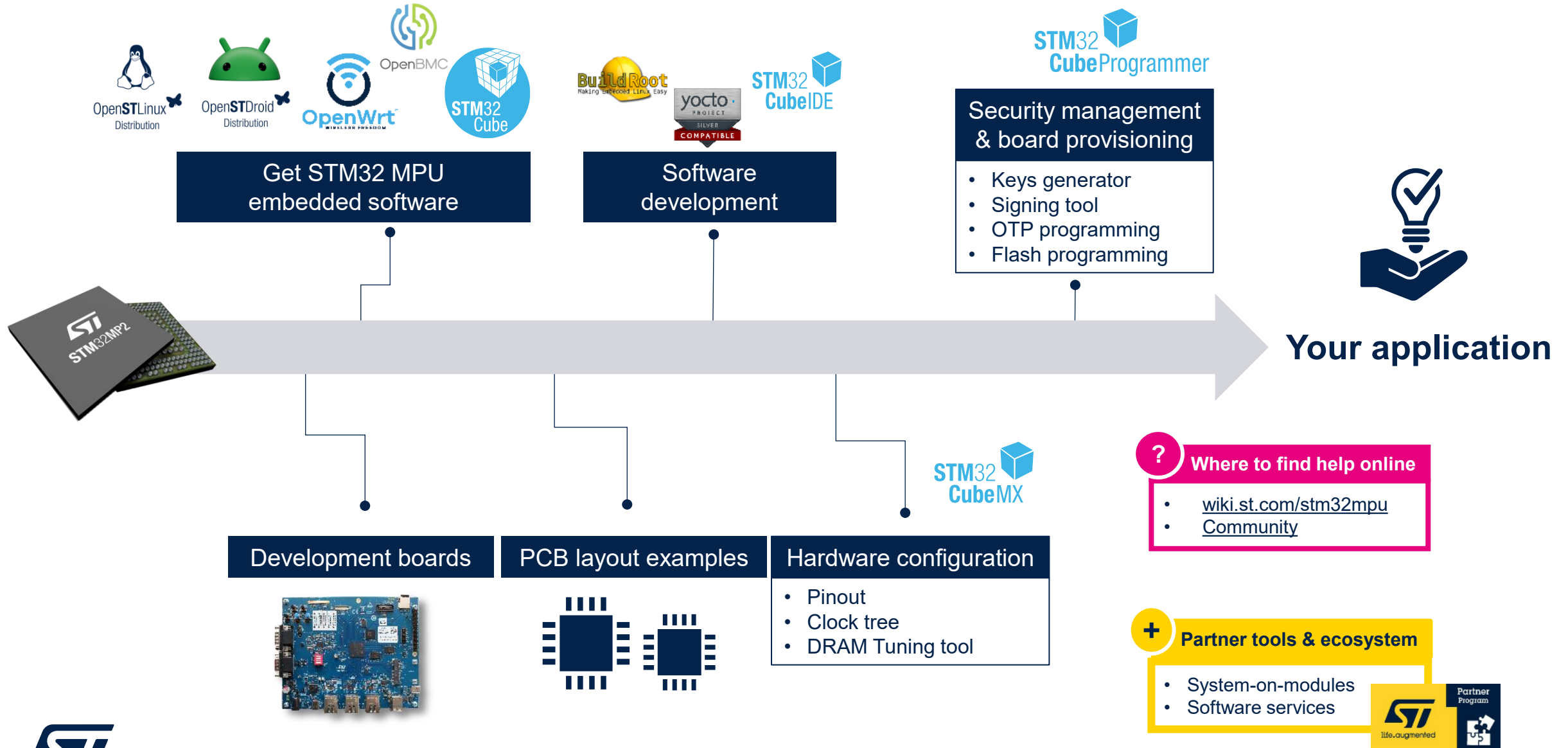
**Reduce development time & cost with
our STM32 ecosystem**



STM32 MPU embedded software: what's new?



Accelerate your time to market



OpenSTDroid Distribution: available only for STM32MP25x product lines

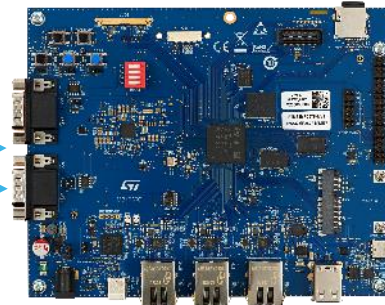


Development tools for the STM32MP2 series

Speed-up evaluation, prototyping, and design



DSI to HDMI
adapter board
B-LCDAD-HDMI1



Evaluation board
STM32MP257F-EV1

MP25	MP23	MP21
☒	☐	☐



LVDS LCD Panel
Display
B-LVDS7-WSVGA



Discovery kit
STM32MP257F-DK

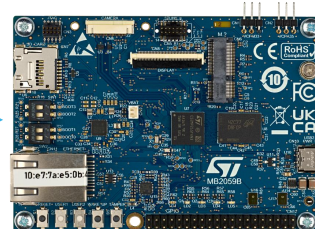
MP25	MP23	MP21
☒	☒	☐



Camera module
adapter board
B-CAMS-IMX



RGB LCD Panel
Display
RK043FN78H-
CT661C



Discovery kit
STM32MP215F-DK

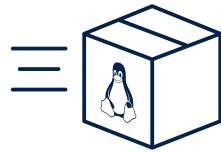
MP25	MP23	MP21
☐	☐	☒



More STM32-based
dev tools available
with our partners

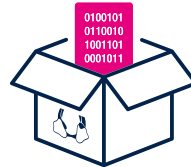
STM32 MPU embedded software

Same Linux software for STM32MP2 series for easy project migration



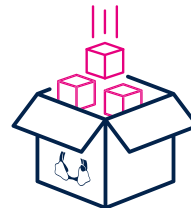
STM32MP2 Starter package

To quickly and easily start with any STM32MP2 microprocessor device



STM32MP2 Developer package

To add your own code on top of the STM32MP2 Embedded Software distribution



STM32MP2 Distribution package

To create your own Linux® distribution as well as your own Starter and Developer packages

[Discover now](#)

STM32 MPU embedded software

Accelerate your time to market using expansion packages

 X-LINUX-ACM	   X-LINUX-IOT	 X-LINUX-SPN1
   X-LINUX-AI	   X-LINUX-MEMS	   X-LINUX-TPM
   X-LINUX-AWS	  X-LINUX-MSP01	 X-LINUX-TSNSWCH
   X-LINUX-AZURE	   X-LINUX-NFC	 X-LINUX-WW4
   X-LINUX-EVG1	 X-LINUX-PREDMNT	  STM32-ISP-IQTune
    X-LINUX-GNSS TBC	  X-LINUX-QT	
 X-LINUX-IGWWSN1	    X-LINUX-RBT1 TBC	
   X-LINUX-IGTW1	    X-LINUX-RT	

Available

Planned

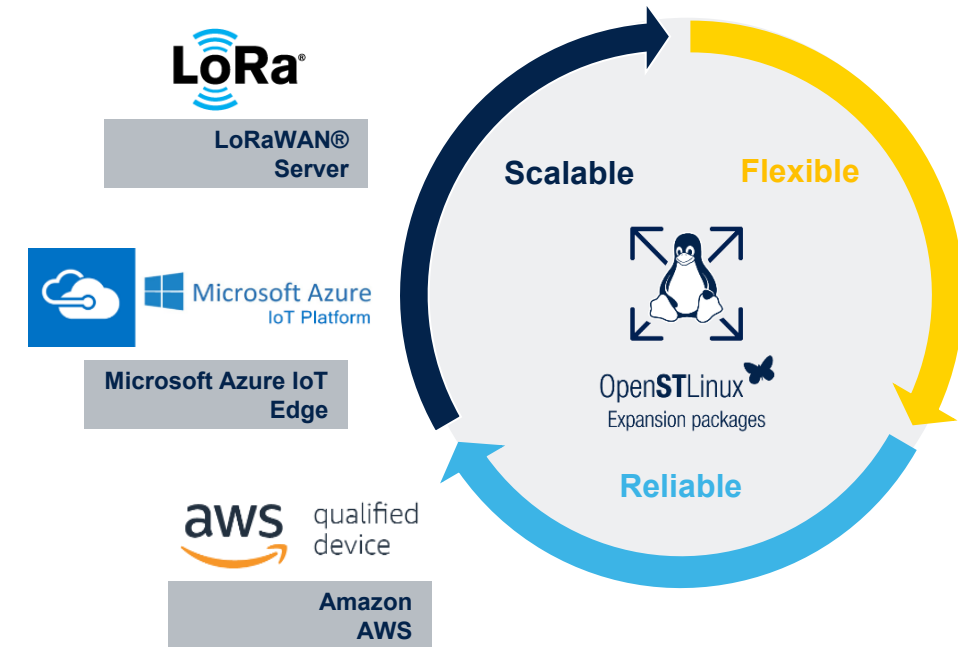
 STM32MP15

 STM32MP13

 STM32MP25

 STM32MP23

 STM32MP21



STM32 MPU OpenSTLinux Expansion Packages



OpenSTLinux long-term Support Releases and support scheme

Extended maintenance to 5 years

2 versions in parallel with
5-year support by ST

References:

- Linux kernel releases: <https://www.kernel.org/category/releases.html>
- Yocto releases: <https://www.yoctoproject.org/development/releases>

OpenSTLinux V5 (Linux 6.1 LTS / Mickledore 4.2)



OpenSTLinux V6 (Linux 6.6 LTS / Scarthgap 5.0 LTS)



OpenSTLinux V7 (Linux LTS / Yocto LTS)



2023 2024 2025 2026 2027 2028 2029 2030 2031



Software development tools

STM32Cube provides the same tools across the STM32MP2 series for greater ease of use

STM32 
CubeMX

STM32 
CubeIDE

STM32 
CubeProgrammer

STM32CubeMX

STM32CubeMX enhanced for MPU

- Device Tree configuration
- Device Tree generation
- DRAM interface tuning tool

IDEs Compile and Debug

Multicore solutions

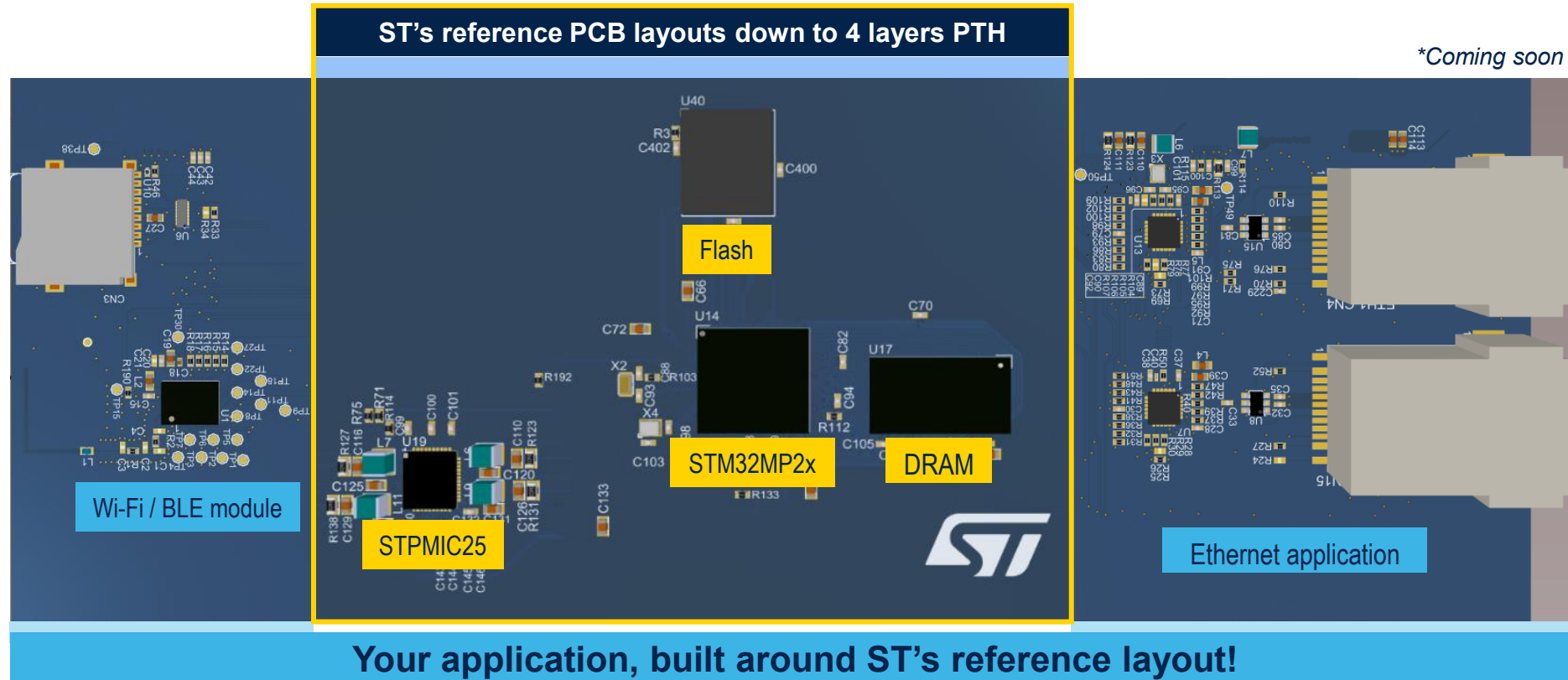
- Free STM32CubeIDE
- OpenSTLinux Developer package support
- Import DRAM tuning project

STM32 programming tool

STM32CubeProgrammer

- Flash, DRAM and/or system memory
- OTP programming
- Signing & key generation tools

Plug & play solution for STM32MP2 series for project reuse



PCB layout examples*
based on Altium projects
provide you with a modular
approach to build your
designs

- All different BGAs packages, STPMIC25, Flash and different DRAM types (DDR3L, DDR4 & LPDDR4)
- Signal integrity and power integrity checks completed
- Developers can reuse the layouts and add their own interfaces linked to their end projects

Speed up your development

Enhance stability, reliability, and simplify your design process with validated DRAM on the STM32 MPU portfolio

Performance optimization

Reduced development costs

Regulatory and quality assurance

[Get the list here](#)

Advanced HMI with stunning graphics

Multiples libraries to run on STM32MP2



Visit our GUI ecosystem webpage

Enhance your added value by relying on ST and Authorized Partner solutions

A growing base of ST Authorized Partners

ST continues to invest in the most recognized open-source standards

From idea to final product, our partners help you build end-to-end solutions

Solutions for edge computing & IoT
from sensors to the cloud

Discover our partners products
and services



Our technology starts with You



Find out more at www.st.com/STM32MP2

© STMicroelectronics - All rights reserved.

ST logo is a trademark or a registered trademark of STMicroelectronics International NV or its affiliates in the EU and/or other countries.

For additional information about ST trademarks, please refer to www.st.com/trademarks.

All other product or service names are the property of their respective owners.



life.augmented